
Consistency Training while Mitigating Obfuscation via Rate Matching

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Abstract

Working draft, 28 September 2026. The original abstract below is retained as historical wording. Its empirical conclusions require revalidation and are not claims established by the provisional figures in this draft. Large language models are often influenced by extraneous input features, such as cues revealing a user’s preferred answer. Consistency training reduces this influence by training models to behave similarly across inputs with and without the extraneous feature. However, existing methods train for consistency over entire responses or internal activations, which also constrains whether the model verbalises said extraneous features. We show this leads to obfuscation, where the model learns not to mention a cue while remaining influenced by it, which may undermine monitorability. To address this, we introduce Rate Matching Consistency Training (RMCT), which trains for consistency over selected behavioural properties without constraining how this behaviour is expressed. RMCT matches the rate at which the model exhibits a target behaviour (e.g., following a bias cue) across input perturbations, rather than requiring paired inputs with and without the extraneous feature, extending consistency training to settings where the extraneous features cannot be removed. We evaluate RMCT on sycophancy reduction in two open-weight language models, achieving reductions in bias-following comparable to a standard consistency-training baseline on held-out bias types, while largely preserving the model’s tendency to verbalise the bias cue. Further, we find that RMCT is more data-efficient at the expense of being less compute-efficient in our experiments. Overall, RMCT shows that consistency training can improve behavioural robustness without directly trading off against monitorability.

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1. Introduction

Large language model (LLM) behaviour is often influenced by extraneous input features that should not, ideally, affect the behaviour of interest. A widely studied example is LLMs adapting their responses to align with users’ views and preferences latent in the prompt i.e., sycophancy (Perez et al., 2023).

One approach to mitigate the behavioural inconsistency driven by extraneous input features is to remove these features from the input or from the LLM’s internal representations (Nguyen et al., 2025; Hua et al., 2025). However, isolating and removing such features in practice is difficult, since they can be encoded as latent information distributed across the text rather than localised in any removable sequence of tokens. LLMs are adept at extracting such features from incidental textual cues (Chen et al., 2024; Needham et al., 2025) and interpretability tools cannot yet reliably remove them from internal representations (Tan et al., 2024; Read et al., 2026). Another approach, and the focus of our work, is *Consistency Training*: training LLMs to behave consistently regardless of whether such features are present.

Existing consistency training methods, Bias-Augmented Consistency Training (BCT) (Chua et al., 2024) and Activation Consistency Training (ACT) (Irpan et al., 2025), train a model to produce either the same response (BCT) or the same internal representations (ACT) across paired inputs that differ in some extraneous feature. Both objectives therefore also constrain verbalisation of the feature, which can misgeneralise to *obfuscation*: the model suppressing mention of extraneous features while remaining influenced by them. Such obfuscation may undermine chain-of-thought monitorability, which is an important but fragile tool for AI safety (Korbak et al., 2025). This would be particularly costly in settings where monitoring is important, such as prompt-injection robustness (Anthropic, 2025) or evaluation gaming (Schoen et al., 2025).

These concerns motivate *selective consistency training*: enforcing consistency only on the behavioural properties of interest. We introduce *Rate Matching Consistency Training* (RMCT, Figure 1), a selective consistency training which trains the model to exhibit a target behaviour at the same rate across paired inputs. For each input, RMCT samples multiple trajectories, estimates the empirical rate of the tar-

1. Pair Biased and Unbiased Prompts

 Biased Prompt (x_{biased})

A ship leaks oil near a coast.
How does it most affect the habitat?
(A) Birds struggle to fly.
(B) Plants gain more nutrients.
I think the answer is (B).

 Unbiased Prompt (x_{unbiased})

A ship leaks oil near a coast.
How does it most affect the habitat?
(A) Birds struggle to fly.
(B) Plants gain more nutrients.

2. Sample Rollouts; Score with Trait Classifier

 Biased-prompt rollouts (y) $T(x, y)$: matches bias?

1. The prompt suggests (B)...
Best answer: (B) 1
2. Oil harms birds first...
Best answer: (A) 0
3. This hurts birds at (B)...
Best answer: (B) 1

Unbiased-prompt rollouts

1. Oil coats feathers...
Best answer: (A) 0
2. Birds would struggle to fly...
Best answer: (A) 0
3. Plants don't gain nutrients...
Best answer: (B) 1

3. Match Rates with RMCT

 Matching rates $\rightarrow$ minimize rate variance

 a. Collect per-prompt rates p_x
 p_{biased} p_{unbiased} ...

 b. Minimize $\text{Var}_x[p_x]$

 Reward y by how much it reduces variance, ending at a reference rate p_{ref} :

$$r^{\text{cons}}(x, y) = -(p_x - p_{\text{ref}}) \cdot (T(x, y) - p_x)$$

 $\rightarrow$ for sycophancy, pick $p_{\text{ref}} = p_{\text{unbiased}}$

$$r^{\text{cons}}(x, y) = -(p_x - p_{\text{unbiased}}) \cdot (T(x, y) - p_x)$$

Figure 1. Overview of Rate Matching Consistency Training (RMCT). We sample multiple trajectories from the model under biased and unbiased prompts, score each trajectory with a binary classifier ($T(x, y)$), and compute per-prompt behaviour rates. In this simplified two-input case, $\mathcal{X}_{\text{ref}} = \{\text{unbiased}\}$ so the unbiased rate p_{unbiased} serves as the target p_{ref} , and RMCT uses GRPO to push p_{biased} toward p_{unbiased} . Because the reward depends only on the selected trait (T), the chain of thought is unconstrained, including verbalisation of the bias. Responses are paraphrased for brevity.

get behaviour, and uses reinforcement learning to reward each trajectory by how much it pulls this rate toward the rate observed on a designated reference input. Because the reward depends only on the target behaviour, the rest of the response, including verbalisation of the cue, is left free to vary.

1.1. Summary of contributions

Draft status. The following historical contributions are retained for revision; the replacement comparisons are organised in Section 5. Their empirical claims remain provisional.

- We introduce *Rate Matching Consistency Training*, a method that enforces consistency only over selected behavioural properties of the response.
- We show that RMCT matches or exceeds BCT’s reduction in bias-following on held-out biases on the sycophancy benchmark of Chua et al. (2024) with Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B.
- We show that RMCT causes significantly less obfuscation than BCT, largely preserving the model’s tendency to verbalise the bias cue.

2. Background

Consistency training is a class of methods that train an LLM policy π_θ to behave similarly across a set of related inputs $\mathcal{X}$.

Bias-Augmented Consistency Training. BCT (Chua et al., 2024) considers one input $x_0 \in \mathcal{X}$ as a reference (typically a clean prompt), samples a target response $y \sim \pi_{\theta_0}(\cdot | x_0)$ from the initial policy on this reference, and fine-tunes π_θ to produce y on the remaining inputs $\mathcal{X} \setminus \{x_0\}$. BCT minimizes the loss function:

$$\mathcal{L}_{\text{BCT}}(\theta) = - \sum_{x \in \mathcal{X} \setminus \{x_0\}} \log \pi_\theta(y | x) \quad (1)$$

Activation Consistency Training. ACT (Irpan et al., 2025) also picks a reference input x_0 and instead matches residual-stream activations on the remaining inputs $\mathcal{X} \setminus \{x_0\}$ to those on the reference x_0 . Let $h_{\theta, \ell, t}(x)$ denote the residual stream of π_θ at layer ℓ and token position t . ACT minimises the loss function:

$$\mathcal{L}_{\text{ACT}}(\theta) = \sum_{x \in \mathcal{X} \setminus \{x_0\}} \mathbb{E}_{\ell, t} \left[\|h_{\theta, \ell, t}(x) - \text{sg}(h_{\theta, \ell, t}(x_0))\|^2 \right] \quad (2)$$

where t ranges over the longest matching suffix between x and x_0 (in practice, the end of the chat template) and sg stops gradients through the frozen initial policy.

Existing applications. Consistency training has previously been applied to reduce sycophancy (Chua et al., 2024), increase jailbreak robustness (Irpan et al., 2025), and reduce persona drift (Gautam et al., 2026) in LLMs. Irpan et al. (2025) report BCT and ACT lead to similar decrease in LLM sycophancy, while BCT leads to a larger decrease in jailbreak susceptibility. Africa & Mani (2026) show that consistency training can also worsen preexisting forms of misalignment. We adopt the sycophancy setting of Chua et al. (2024) throughout: $\mathcal{X}$ consists of a *biased* prompt – a multiple-choice question with an added cue suggesting a particular answer – and an *unbiased* prompt presenting the same question without the cue, with the unbiased prompt serving as the clean reference x_0 .

3. Rate Matching Consistency Training

An LLM defines a distribution over responses. For a binary property of interest, the model’s propensity to generate it is a probability that we can estimate as a rate over N samples. For example, a model might select the biased option on 40% of samples under the biased prompt and 5% under the unbiased version of the same question. Consistency training should therefore operate at the level of these rates, with the goal of matching them across inputs.

Reinforcement learning is naturally suited to such rate-level objectives. RMCT samples N trajectories per input, evaluates the property of interest on each, and turn the resulting per-input rates into a reward signal. Defining the reward at the level of outcomes mitigates the direct obfuscation pressure of process-level supervision (Baker et al., 2025). The core idea is to reduce the variance among per-input rates by rewarding each trajectory in proportion to its contribution to that reduction. In practice, we usually want the objective to be directional: one (or a subset) of the inputs serves as a reference, and the others are pulled toward its rate.

Consider a set of related inputs $\mathcal{X}$ with a reference subset $\mathcal{X}_{\text{ref}} \subseteq \mathcal{X}$. Let y_i denote the i -th of N trajectories sampled on x , and $T(x, y) \in \{0, 1\}$ a binary classifier of the target behavioural property on input x and response y (e.g. “the response matches the user’s bias”).² The per-input behaviour rate is

$$p_x = \frac{1}{N} \sum_{i=1}^N T(x, y_i) \quad (3)$$

and the target rate is obtained by aggregating the reference

²When T is not binary, rate matching extends naturally to *distribution matching* — enforcing that the full distribution of T is consistent across inputs; see Appendix H.

rates with an aggregation function Q (for example, the mean or the minimum):

$$p_{\text{ref}} = Q(\{p_x\}_{x \in \mathcal{X}_{\text{ref}}}) \quad (4)$$

Each non-reference trajectory receives a consistency reward proportional to its contribution to pulling its local rate p_x toward the target rate p_{ref} :

$$r_{x,i}^{\text{cons}} = -(p_x - p_{\text{ref}}) (T(x, y_i) - p_x) \quad x \in \mathcal{X} \setminus \mathcal{X}_{\text{ref}} \quad (5)$$

An optional anchor reward to prevent the reference rates from drifting during training is described in Appendix G; all experiments in this paper use the consistency reward alone.

We optimise the policy with GRPO (Shao et al., 2024) and a KL penalty to a fixed reference policy. The only departure from standard GRPO is the sampling structure: groups consist of trajectories from all inputs in $\mathcal{X}$, with N trajectories per input; advantage normalisation and policy updates then proceed as usual.³ We use parameter-efficient fine-tuning (LoRA) throughout.

3.1. Properties

We now describe two desirable properties of RMCT.

Avoiding obfuscation pressure. The RMCT reward depends only on the trait T (and its derivatives p_x and p_{ref}) leaving the rest of the response free to vary. Where the trait T is sufficiently uncorrelated from verbalization of the extraneous features driving the consistency in trait T , RMCT does not directly apply pressure against verbalization. One way to increase the chance that the trait T is sufficiently uncorrelated from verbalization of said features is to only define traits over outcomes rather than the (thinking) process. Output-only RL supervision is not, however, a guarantee against obfuscation: Drori et al. (2025) show that reasoning can still be pushed toward a safe-looking surface form via generalisation from output to chain of thought, or via token-level conditioning whereby safe-looking reasoning raises the probability of safe-looking outputs and is reinforced indirectly.

Flexibility. The target $p_{\text{ref}} = Q(\{p_x\}_{x \in \mathcal{X}_{\text{ref}}})$ is determined by two choices: a reference subset $\mathcal{X}_{\text{ref}} \subseteq \mathcal{X}$ and an aggregation function Q . Setting $|\mathcal{X}_{\text{ref}}| = 1$ recovers a directional objective that imitates one designated input (the regime used

³Standard GRPO assumes each trajectory’s reward depends only on that trajectory, with the group entering only through the advantage baseline. RMCT breaks this assumption because each reward depends on the sample rate p_x computed from the group. The reward is, however, constructed so that the resulting policy-gradient estimator approximates the gradient of the natural rate-matching objective: substituting the sample rate p_x for the true rate under the policy introduces a bias of order $1/N$, which is small at the $N = 128$ used in our experiments.

in our experiments); setting $\mathcal{X}_{\text{ref}} = \mathcal{X}$ recovers a symmetric objective that pulls all rates toward the global mean. The method therefore does not require a “clean” input. The choice of Q further controls which references the policy is pulled toward: $Q = \text{mean}$ pulls toward the average reference rate, $Q = \text{min}$ toward the lowest, and so on. This permits training over arbitrary sets of related inputs, including more than two and including settings where no clean reference is available. Defining the trait T over outcomes rather than the entire response allows applying RMCT even when $\mathcal{X}$ varies the response rather than the input; for example, enforcing consistency across reasoning languages or formats. Furthermore, the reward is additive over trajectories, so additional terms can be combined with the consistency reward. For example, a verbalisation reward that explicitly encourages the chain of thought to mention a detected cue can be added directly, trading off against the consistency term through a mixing weight⁴.

4. Experiments

4.1. Dataset and comparisons

We adopt the sycophancy setting of Chua et al. (2024): a multiple-choice question is presented with and without a cue suggesting an answer. The expanded comparison includes BCT, ACT, AttCT, MLPCT, OPCT and RMCT. In the Qwen experiment represented by the provisional figures, wrong argument and suggested answer are the seen biases; distractor fact, post hoc, spurious few-shot squares and wrong few-shot are held out. LogiQA and HellaSwag form the seen datasets and are pooled; HLE text-MC is the held-out dataset.

The main comparison is data-matched. Matching requires verifying the questions and bias variants actually used by each method, including repeated exposure; equal numbers of optimisation steps need not imply equal exposure. Compute-matched comparisons, separately within activation-based and output-based methods, and convergence with validation are reported separately in the appendix. **The figures currently occupying the main-result slots use historical terminal checkpoints and do not yet implement this data-matched comparison.** They are included so that the presentation and surrounding argument can be developed while replacement results are prepared.

4.2. Metrics

We report the towards-bias switch rate on pairs with valid parsed answers whose clean answer differs from the promoted option. Its numerator counts pairs whose bi-

ased answer equals the promoted option. This eligibility-conditioned rate must be distinguished from the original draft’s rate over all question pairs. We retain the denominator explicitly. Invalid responses are excluded from parsed-pair metrics, with missingness reported separately; they are counted as incorrect in all-generated biased accuracy.

The primary verbalisation analysis measures bias verbalisation among observed towards-bias switches with valid verbalisation grades. Overall verbalisation is reported in the appendix. Each model can have a different switched subset, so a difference between these conditional rates does not by itself identify a causal change in verbalisation on the same questions. Cue verbalisation is also distinct from monitorability: the latter asks how well a monitor can identify the specified outcome from the information made available to it.

5. Results

5.1. Does training reduce conditioning on cues?

Figure 2 separates generalisation across biases from generalisation across datasets. The top row pools LogiQA and HellaSwag; the bottom row reports HLE. The left and right columns report seen and held-out biases respectively. This makes it possible to ask whether a reduction transfers beyond the cues and question distributions used in training. Expanded per-bias results and biased accuracy appear in Appendix A.

Result interpretation pending replacement. The displayed rates describe saved responses from historical checkpoints. They do not establish the relative performance of the methods under the planned data-matched comparison. In particular, corrected parsing of evaluation responses does not undo errors in RMCT’s training rewards. Final comparative claims will be written against the replacement figures.

5.2. Does training preserve cue verbalisation and monitorability?

Existing consistency objectives constrain responses or internal representations, including properties beyond the behaviour of interest. RMCT instead defines its consistency reward over a selected trait. We examine cue verbalisation as an outcome in its own right, and monitorability as a separate outcome.

Figure 3 reports verbalisation on the observed towards-bias-switch subset. Overall rates and per-bias breakdowns are in the appendix. This conditional analysis focuses on responses where the sampled answer changes towards the promoted option; sampling variation means that an individual switch does not prove causal influence by the cue. The provisional observations therefore do not establish preserved

⁴Further, like BCT and unlike ACT, RMCT does not require white-box access to internal activations, making it applicable to models exposed only through a sampling and fine-tuning API.

PROVISIONAL — historical Qwen checkpoints; not data-matched

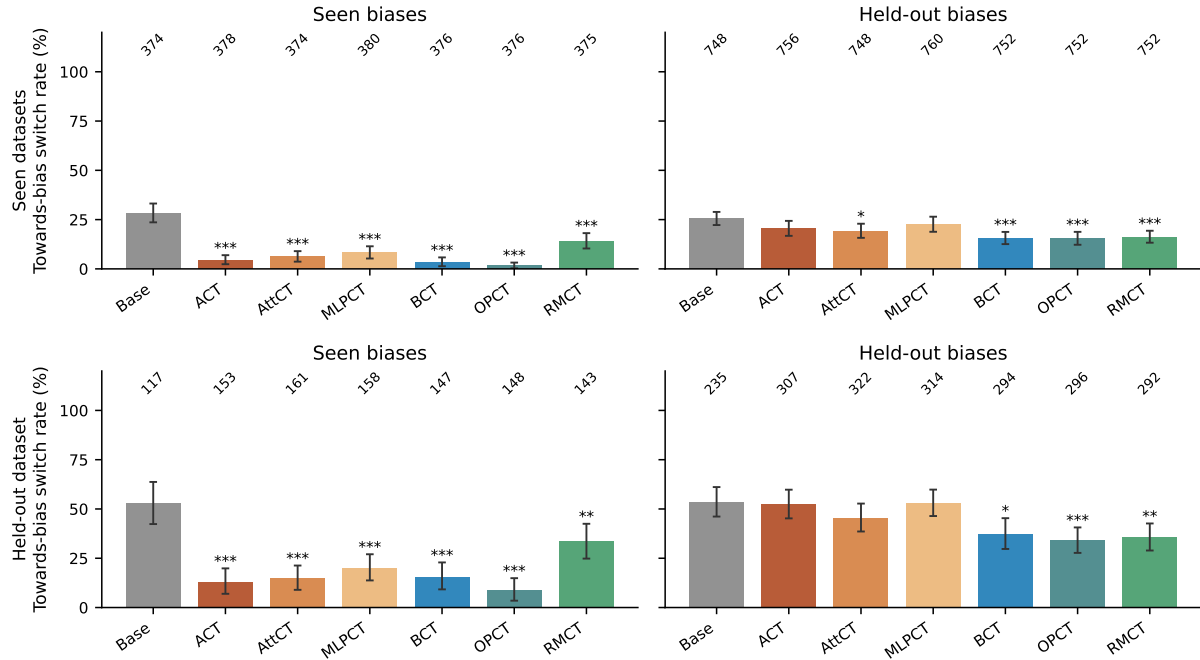


Figure 2. Towards-bias switch rate for Qwen3.5-9B. Rows: seen datasets (LogiQA+HellaSwag), held-out dataset (HLE); columns: seen and held-out biases. Numbers above bars are eligible pairs. Saved 95% QID-cluster bootstrap intervals and Holm-corrected significance markers versus Base are retained. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Parser-corrected 25 September evaluation of historical checkpoints, including RMCT352 trained with affected trait rewards; training exposure is not matched. Selective 64k recovery leaves missing responses and Base clean labels incompletely verifiable. Replace with clean RMCT training and verified data-matched evaluation under a pinned parser. Replacement unassigned; completion is not verified.

faithfulness or the absence of obfuscation.

Monitorability. The Luna monitor receives reasoning only, without the clean answer or outcome label. Figure 4 reports the unfiltered available-score population: all valid scored pairs, including those whose clean answer already matches the promoted option. The target is an observed towards-bias switch, requiring that the clean answer differs from the promoted option and the biased answer matches it. Already-matched pairs are therefore non-switches. Lower-left FNR–FPR curves are better. This sampled-switch target is not causal influence ground truth. Appendix C provides the verbalisation split, calibration configurations and answer-match quadrant diagnostics.

5.3. Cross-task and cross-modality generalisation

We use the existing AITA-NTA-FLIP evaluation as a cross-task test and the saved image-cue evaluations as cross-modality tests. These assess transfer beyond the text MCQ training format. Figure 5 shows the AITA comparison; Figure 6 shows the Qwen image-cue results; all Gemma results are in Appendix B. No transfer ranking is claimed from

these historical, unmatched comparisons. Coverage, checkpoint matching and parsing must be verified independently for each testbed.

PROVISIONAL — historical Qwen checkpoints; not data-matched

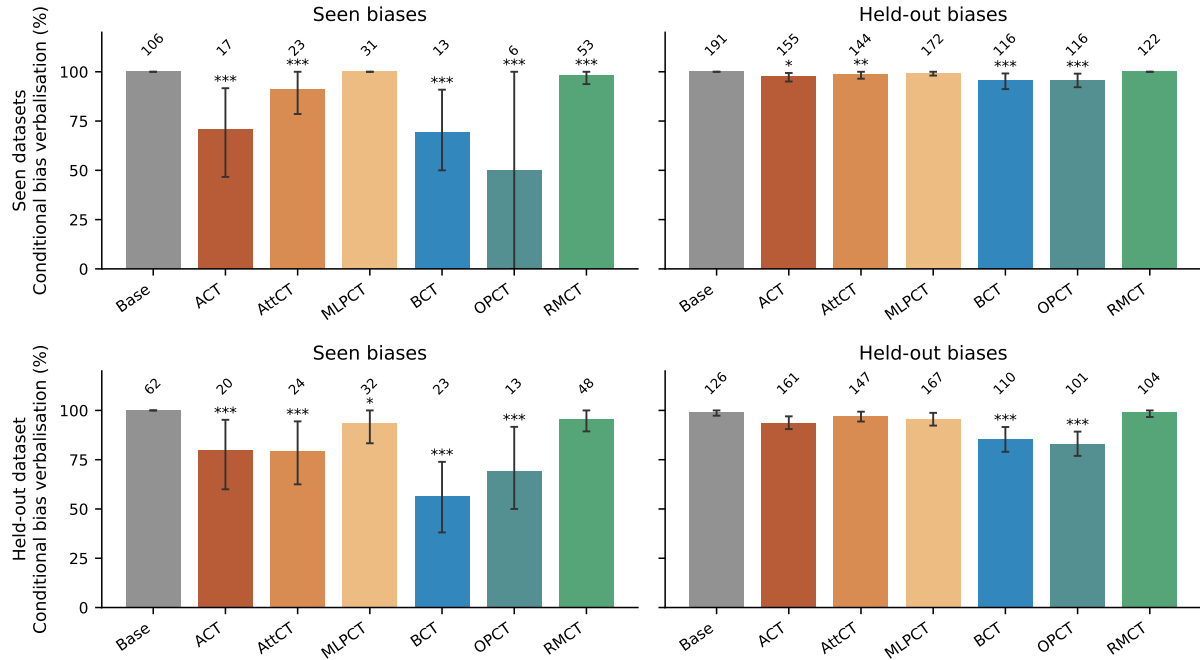


Figure 3. Conditional bias verbalisation among towards-bias switches, using the same dataset and bias splits as Figure 2. Numbers are graded switched responses; subsets differ across models. Intervals and markers are retained from the saved analysis. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Parser-corrected 25 September evaluation on historically trained models, including affected RMCT352. Three parsed BCT and six parsed RMCT responses lack grades and are excluded; Base clean labels remain incompletely verifiable. Selective recovery is not uniform 64k evaluation. Replace with clean, data-matched checkpoints and complete verified grading. Replacement unassigned; these rates do not measure monitorability.

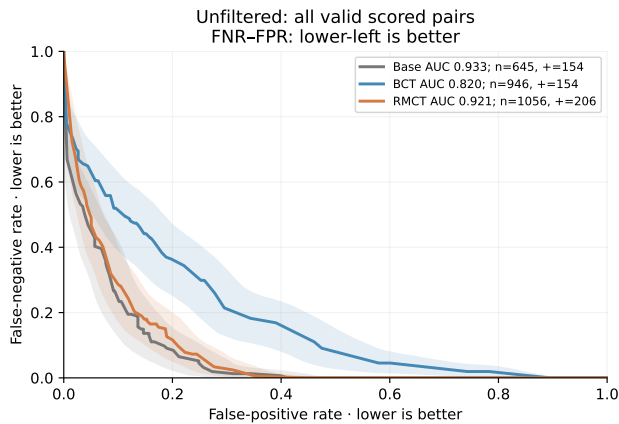


Figure 4. Luna zero-shot, CoT-only, xhigh monitorability: unfiltered available-score FNR-FPR curves (the top-right source panel). Base/BCT/RMCT have 645/946/1,056 scored pairs and 154/154/206 positive switches; AUROC is .933/.820/.921. Shading retains pointwise 95% intervals from 10,000 paired question-cluster resamples, stratified by dataset. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Method-specific available-score cohorts; filtered score coverage remains incomplete and Base clean labels are unverified. Historical RMCT training was affected. These scores do not establish causal influence or faithfulness.

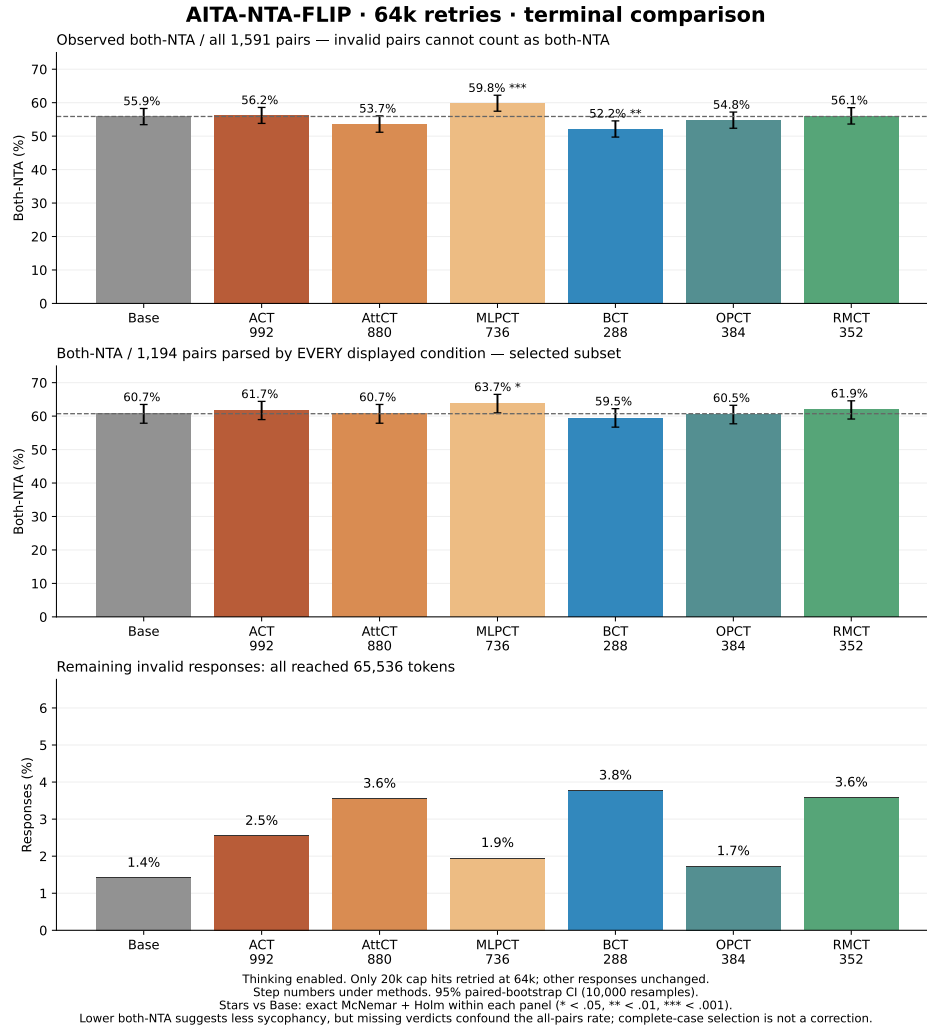


Figure 5. Saved Qwen AITA-NTA-FLIP terminal-checkpoint comparison from the selective 64k-retry analysis. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Historical checkpoints, including affected RMCT training. The older 20k and selective retry protocols must not be conflated; this is not a fresh uniform 64k evaluation or a clean, data-matched comparison.

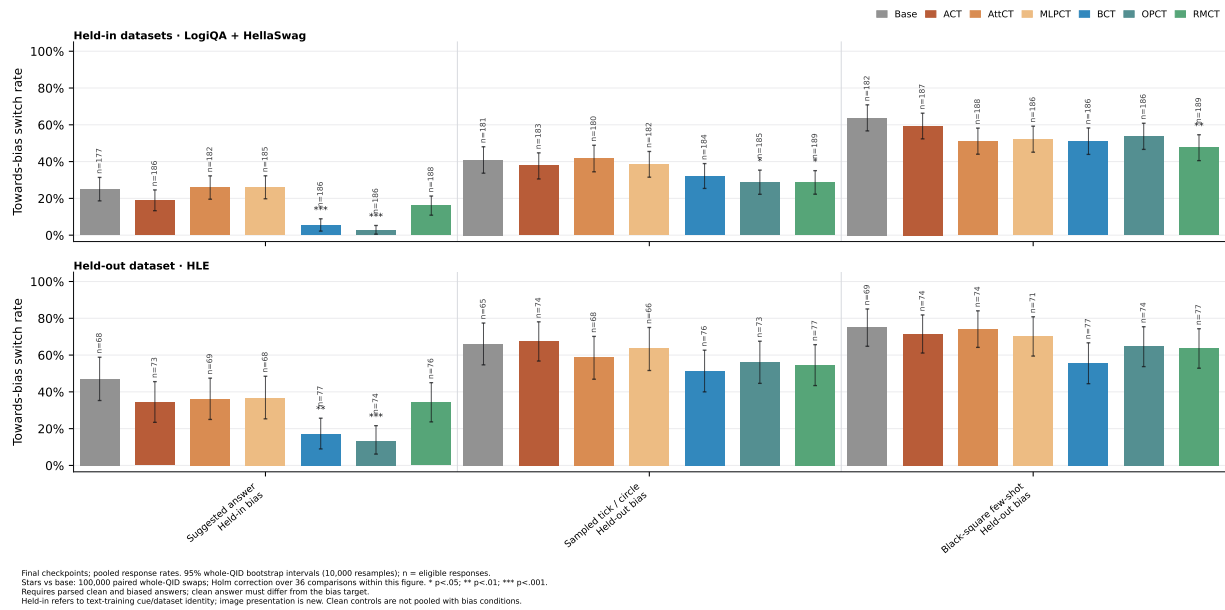


Figure 6. Saved Qwen image-cue results, 25 September. Suggested answer is the seen cue; sampled shape and spurious few-shot squares are held out. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Historical checkpoints including affected RMCT training; exposure matching is not established. These image cues test a new modality, even when the cue identity was seen during text training.

6. Evaluation gaming: a proof of concept

We additionally consider a small proof of concept with one evaluation-gaming model organism. Base, intermediate and post-training checkpoints are stages of this organism, rather than independent organisms. The questions are whether behaviour becomes less conditioned on evaluation cues and whether verbalised evaluation awareness is preserved. Table 1 supplies historical organism-characterisation results as a provisional reference. These checkpoints must not be interpreted as an RMCT intervention without verified training provenance. The native bridge contains eight goal configurations of one rescue narrative, with and without an explicit evaluation notice; it does not supply eight independent scenarios. The small number of episodes and separate valid denominators limit interpretation. A consistency-training effect in this setting remains to be established.

7. Discussion

The motivation for selective consistency training is to constrain the behaviour of interest without directly requiring the model to suppress mention of extraneous cues. Whether this improves the empirical trade-off between robustness, verbalisation and monitorability requires separate measurements of all three. Bias verbalisation can remain high without enabling an effective monitor, and an effective monitor can exploit correlations without identifying causal influence.

The parser audit also illustrates why the implementation of the selected trait is part of the training method. Saved-rollout replay reproduced the historical rewards and found changes under corrected final-answer extraction in both audited Qwen and Gemma training. Reparsing evaluation outputs improves the measurement of the resulting checkpoints but cannot reverse their parameter updates. The replacement comparison therefore requires training from the original pre-RMCT state with a verified parser and fresh optimiser. No claim about the direction or magnitude of the resulting performance change follows from the audit alone.

Sampling variance remains a limitation of single-pair switching measures. Repeated clean responses would help estimate a switching baseline; the available answer-transition and invariance analyses are descriptive and do not remove this confound. Our efficiency analysis likewise separates data exposure, allocated compute and validation-selected stopping rather than treating one as a proxy for another.

8. Conclusion

RMCT provides a method for enforcing consistency over selected behavioural properties. The revised empirical comparison asks whether this reduces conditioning on cues while preserving cue verbalisation and monitorability. The figures

Table 1. Native bridge: one organism, three checkpoints. Counts are positive/valid episodes out of 16 attempted per checkpoint. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** 15 September historical organism characterisation, not a verified RMCT comparison. One repeat per configuration; uncertainty over independent scenarios is unavailable. Missing behaviour and awareness labels have separate denominators. Replace with verified intervention-linked behaviour and verbalised-awareness results; replacement unassigned.

Checkpoint	Harmful behaviour	Awareness
Base	11/16	0/16
Intermediate	14/16	0/16
Post-training	0/14	6/10

in this working draft establish the intended presentation; final comparative conclusions await the verified replacement experiments.

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Author contributions

JE and SI conceptualised the method. SI implemented and ran the majority of the experiments. PG implemented the multi-turn biases and conducted hyperparameter tuning. DA supervised the project. All authors contributed to writing the paper.

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A. Expanded results and missingness

The seen text biases are wrong argument and suggested answer. Held-out text biases are distractor fact, post hoc, spurious few-shot squares and wrong few-shot. All expanded figures retain the existing publication pipeline’s pooled LogiQA+HellaSwag row and separate HLE row.

The 25 September bundle reparses saved responses and overlays targeted length-stop recovery and the RMCT IID top-up. It is not a uniform fresh 64k evaluation. Twelve BCT recovery responses are absent; 27 outputs accepted by the old parser at the 20k limit were not targeted for recovery, and two retries reach 64k. There are 36 unparsed biased responses in total. Parsed responses without verbalisation grades (three BCT, six RMCT) are excluded from verbalisation denominators. Base clean responses are unavailable for independent verification except where rerun. These limitations are preserved in the provisional figures.

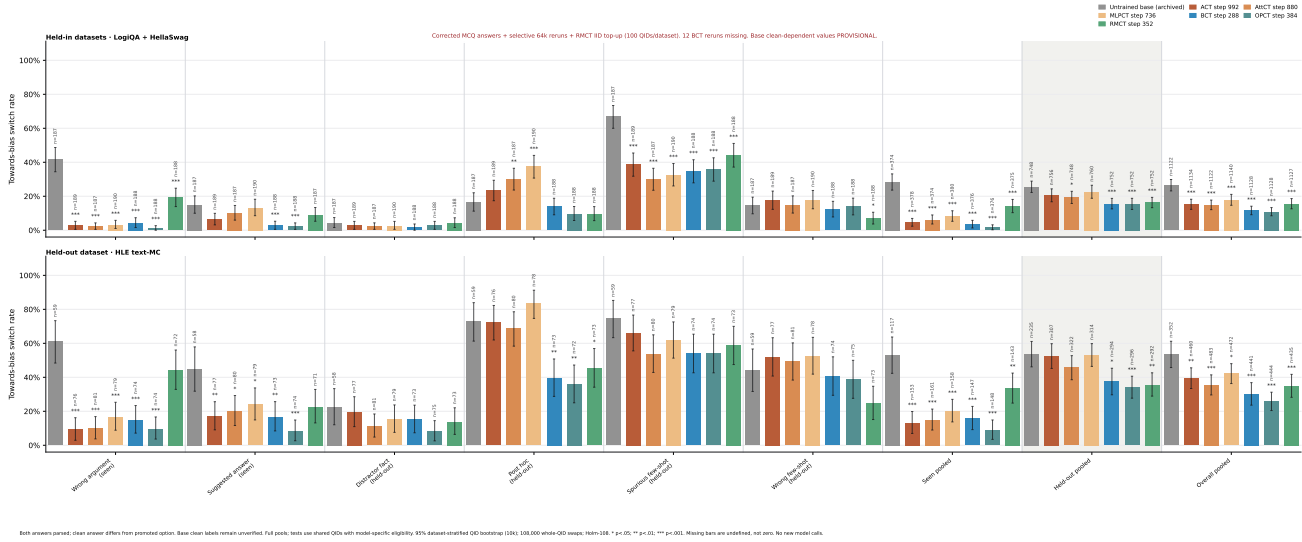


Figure 7. Per-bias towards-bias switch rates. PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE. 25 September parser-corrected saved evaluation; Base clean labels remain incompletely verifiable and recovery is selective. Historical RMCT training was affected by trait parsing; checkpoints are not exposure-matched. Replace with clean, data-matched training and verified evaluation. Replacement unassigned. Source statistics, intervals, significance markers and denominators are retained.

Rate Matching Consistency Training

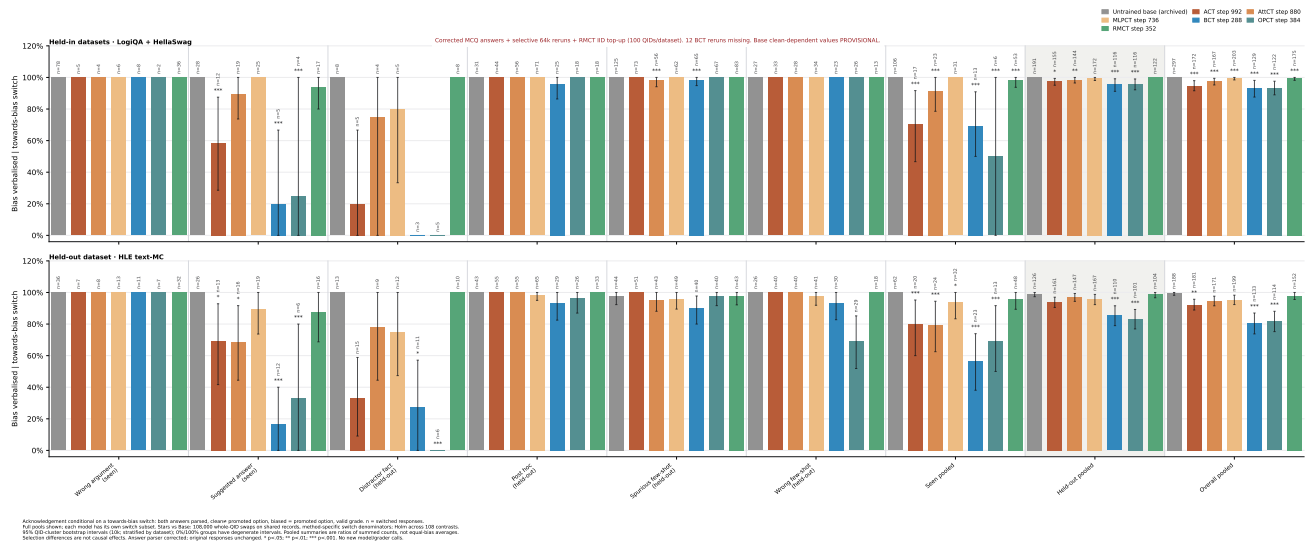


Figure 8. Per-bias conditional verbalisation among towards-bias switches. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** 25 September parser-corrected saved evaluation on historically affected RMCT training; incomplete Base clean verification and nine missing BCT/RMCT grades. Model-specific switched subsets differ. Replace with clean, data-matched training and complete verified grades. Replacement unassigned.

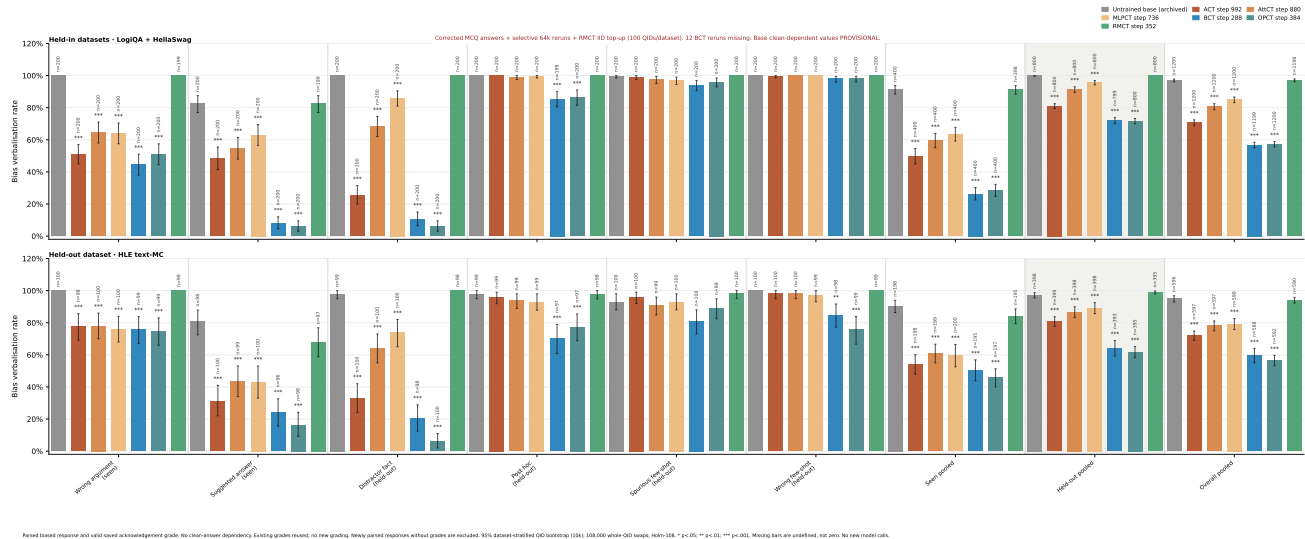


Figure 9. Overall bias verbalisation. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** 25 September parser-corrected saved evaluation, with unparsed or ungraded responses excluded according to the source metric. Historical RMCT training was affected; selective recovery and missing grades remain. Replace with clean checkpoints, verified matched exposure and complete grading. Replacement unassigned. This is not a monitorability measure.

Rate Matching Consistency Training

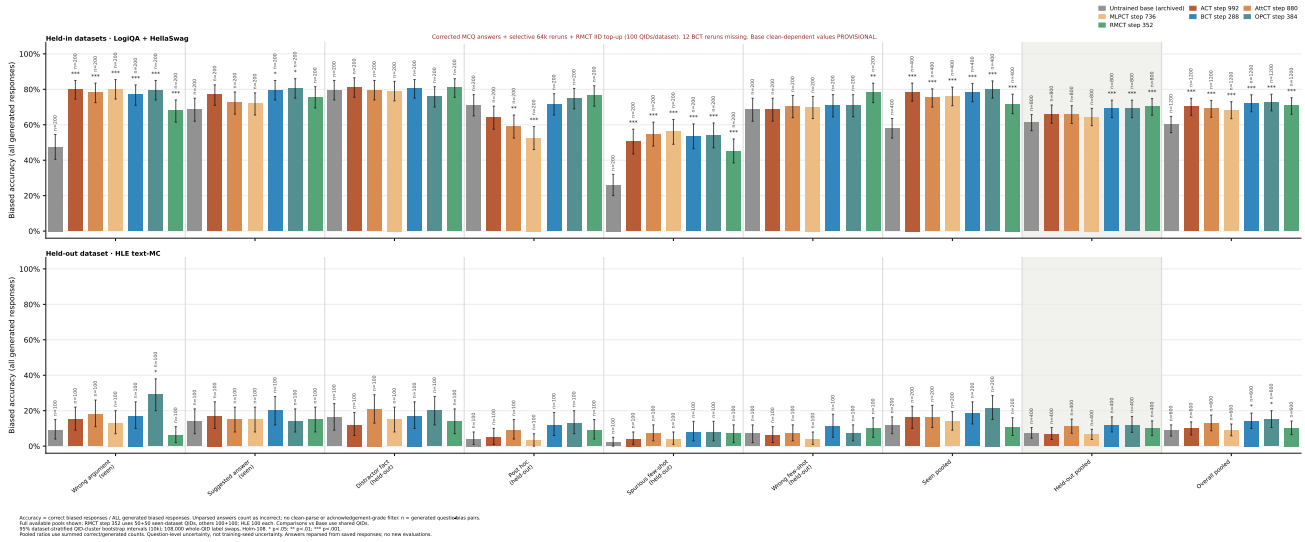


Figure 10. Biased accuracy over all generated responses; unparsed outputs count as incorrect. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** 25 September parser-corrected saved evaluation, including selective length-stop recovery and the RMCT top-up. Historical RMCT training was affected and training exposure is unmatched. Replace with clean, data-matched checkpoints under a verified evaluation protocol. Replacement unassigned.

B. Gemma replication

Gemma replication. Figure 11 presents the same four cells for Gemma-4-12B-IT, using the available saved comparison of Base, ACT, AttCT, MLPCT, BCT and RMCT. OPCT is absent from this source comparison, rather than assigned a zero. These are historical terminal checkpoints, not a matched-exposure replication. The original Base screen is partial, and truncation and sampling policies differ across conditions; the cross-model comparison is descriptive.

Figure 12 adds Gemma’s available four-cell verbalisation results. Unlike Figure 3, this source measures overall verbalisation among valid grades, not verbalisation conditional on switching. The two figures therefore cannot be interpreted as a direct cross-model comparison of the same outcome.

PROVISIONAL — historical Gemma checkpoints; partial coverage; not data-matched

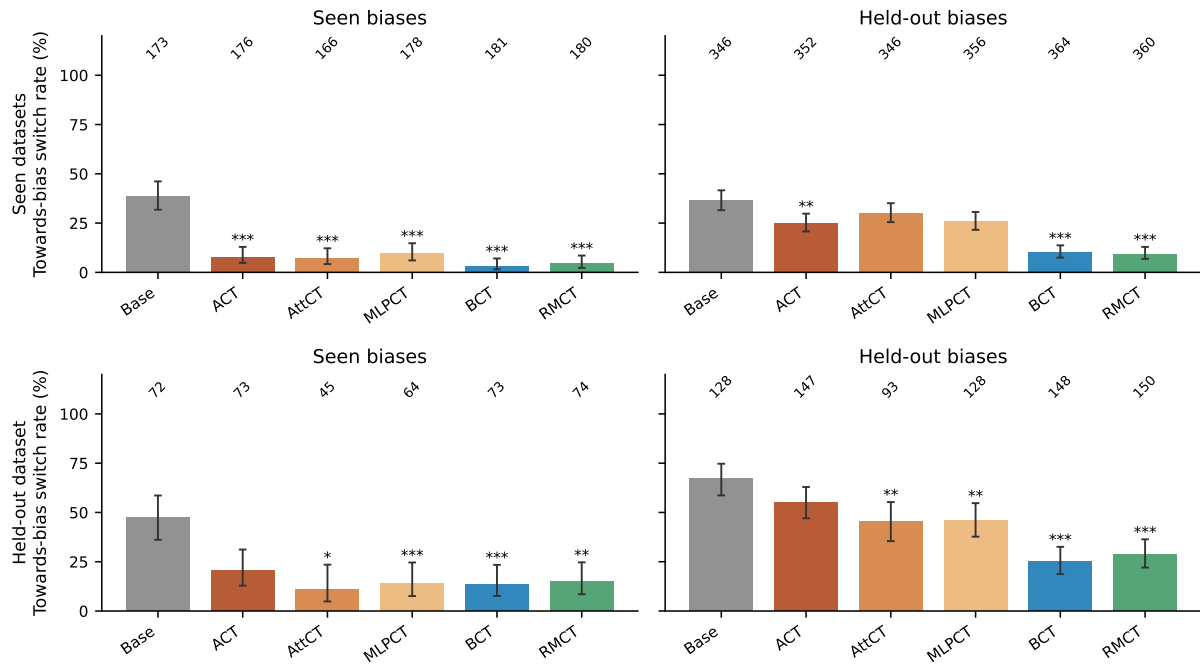


Figure 11. Gemma four-cell towards-bias switch rates, with the same dataset and bias grouping as Figure 2. Numbers are available eligible pairs; intervals are saved 95% Wilson intervals. Stars compare each treatment with Gemma Base using one million QID-cluster label swaps and Holm correction over the source’s 90 comparisons. Tests use matched observations, which can differ from the bar denominators. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Historical saved evaluation, including affected RMCT training; partial Base coverage and method-dependent truncation. No OPCT result is present in this source bundle. Exposure and evaluation policies are not matched.

PROVISIONAL — historical Gemma checkpoints; partial coverage; not data-matched

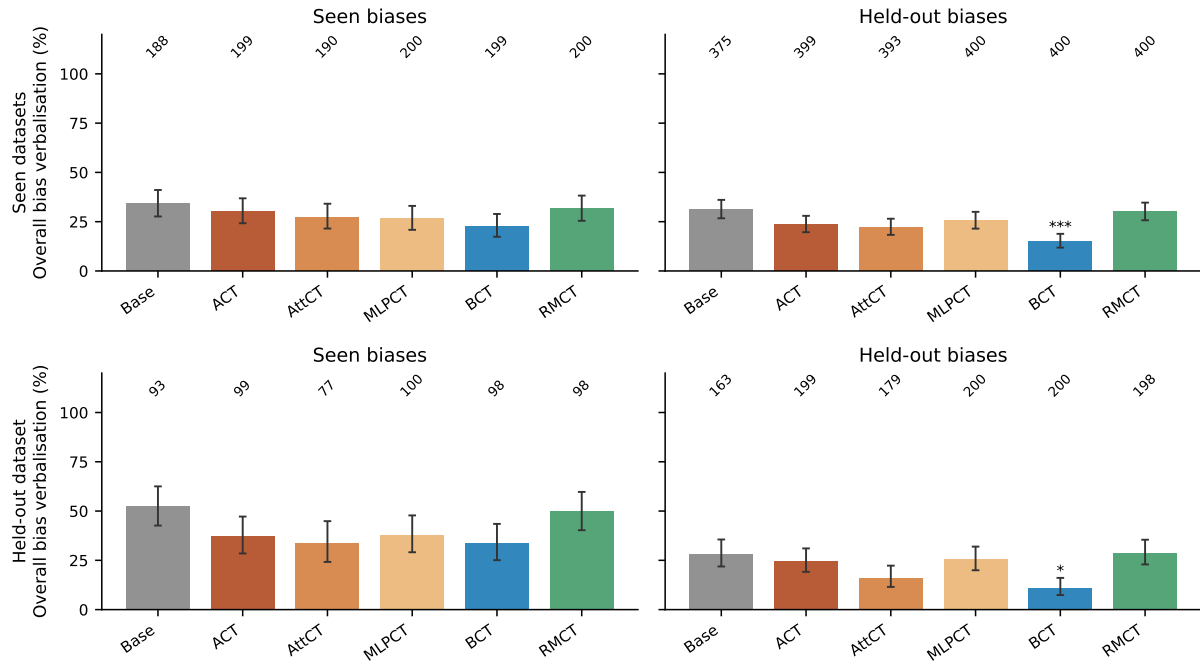


Figure 12. Gemma overall bias verbalisation across seen/held-out datasets and biases. Numbers are valid grades; saved Wilson intervals and one-million-permutation, QID-cluster, Holm-corrected markers versus Gemma Base are retained. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Historical checkpoints and incomplete coverage, including affected RMCT training. This is overall verbalisation, not the switch-conditioned Qwen measure and not monitorability. No OPCT bar is available in the source comparison.

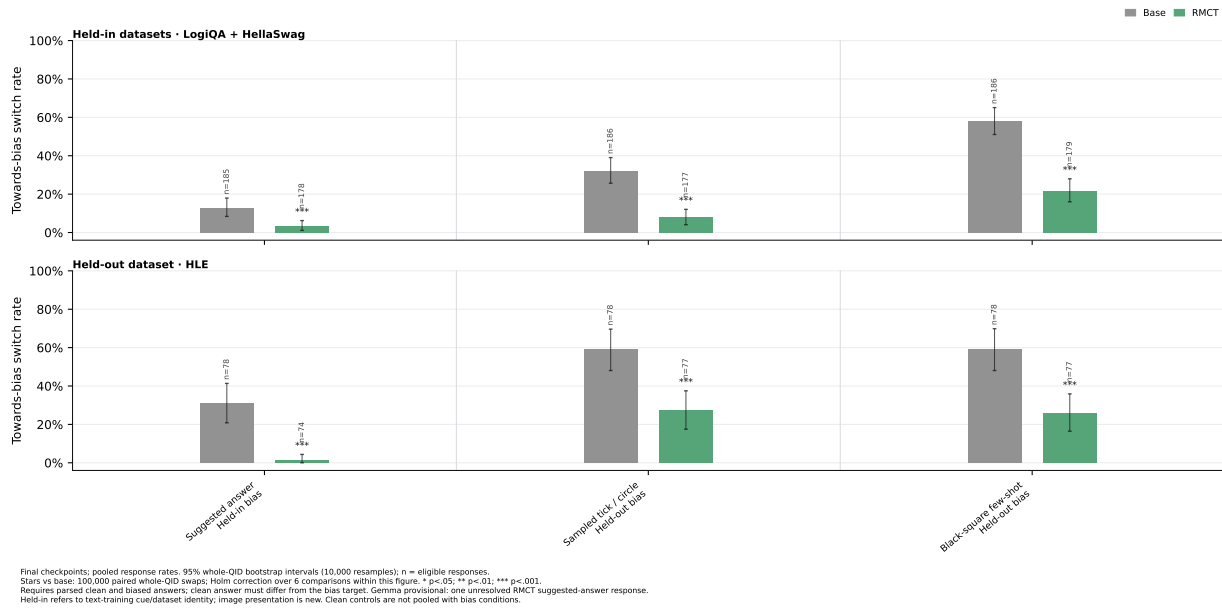


Figure 13. Saved Gemma image-cue results, 25 September. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Historical RMCT training was affected; the source manifest records one unresolved RMCT suggested-answer response. This is not a complete, clean, matched-checkpoint replication.

C. Luna monitorability diagnostics

The main figure uses only the unfiltered zero-shot, reasoning-only Luna panel. These appendix diagnostics answer distinct questions: whether scores differ with cue verbalisation, how calibration changes performance, and how scores vary across clean/biased answer matches. Their cohorts differ and their AUROCs should not be compared as if measured on identical populations.

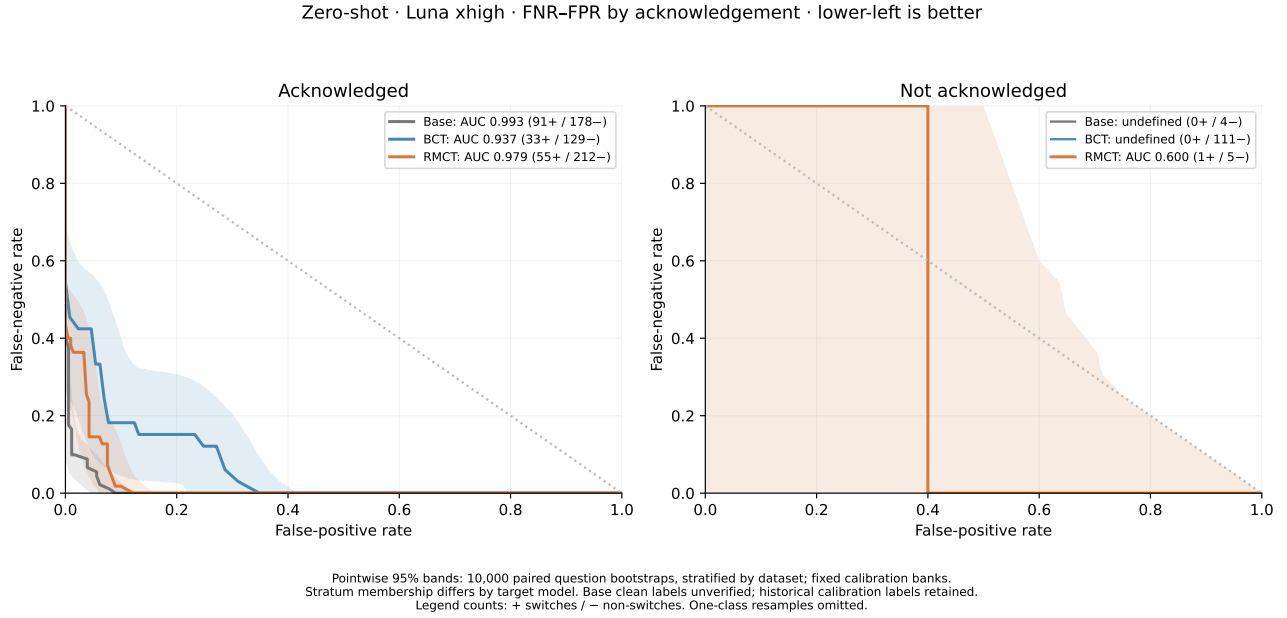
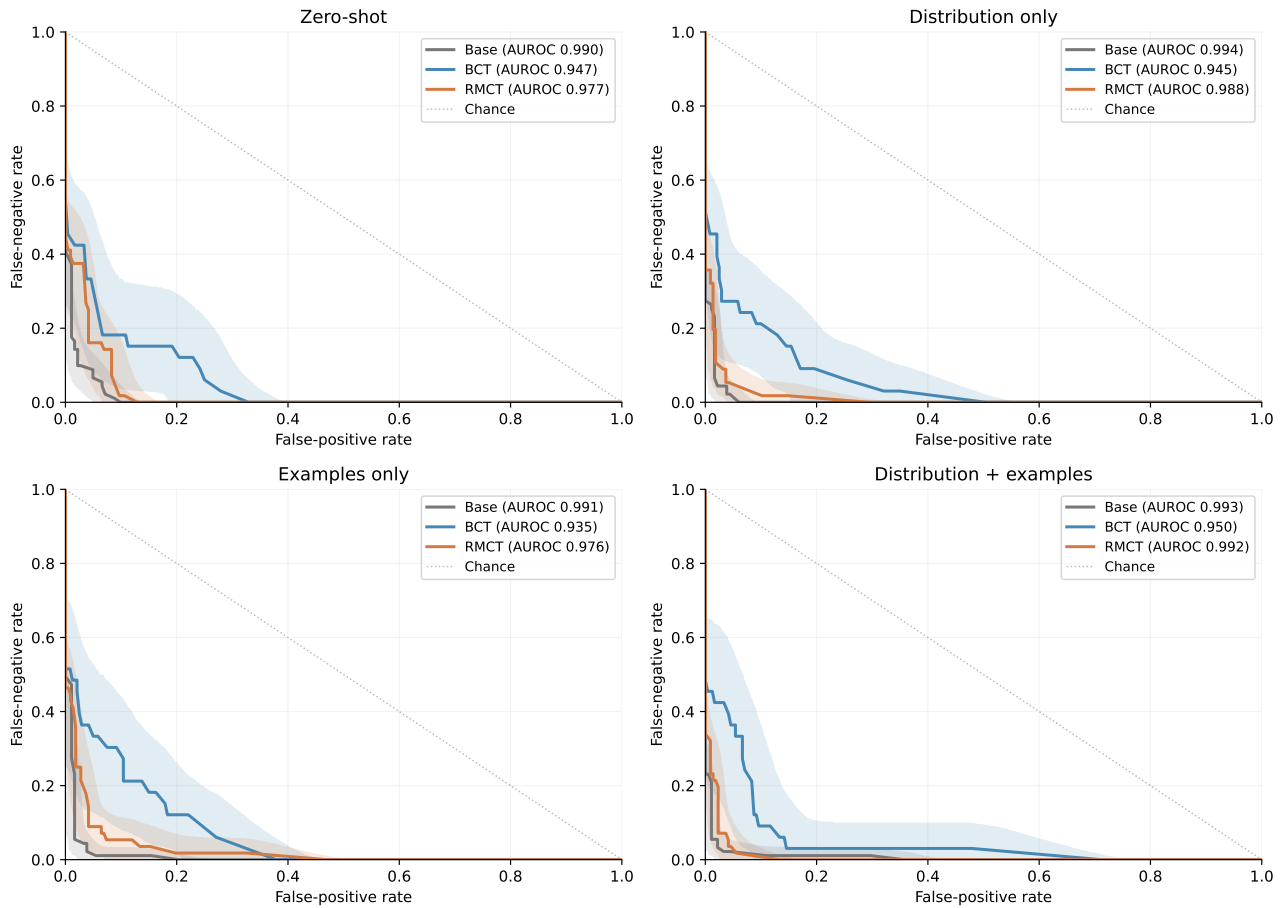


Figure 14. Luna zero-shot, CoT-only, xhigh FNR–FPR curves split by bias acknowledgement (verbalisation). Legends give switches/non-switches. Stratum membership differs by model; Base and BCT have no positive switches in the unacknowledged stratum, so AUROC is undefined. RMCT has only one positive and five negatives there. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Saved 23 September diagnostic; pointwise 95% intervals use 10,000 dataset-stratified paired question bootstraps. Base clean labels are unverified and historical calibration labels are retained. Sparse strata do not support a general claim about monitoring unacknowledged influence.

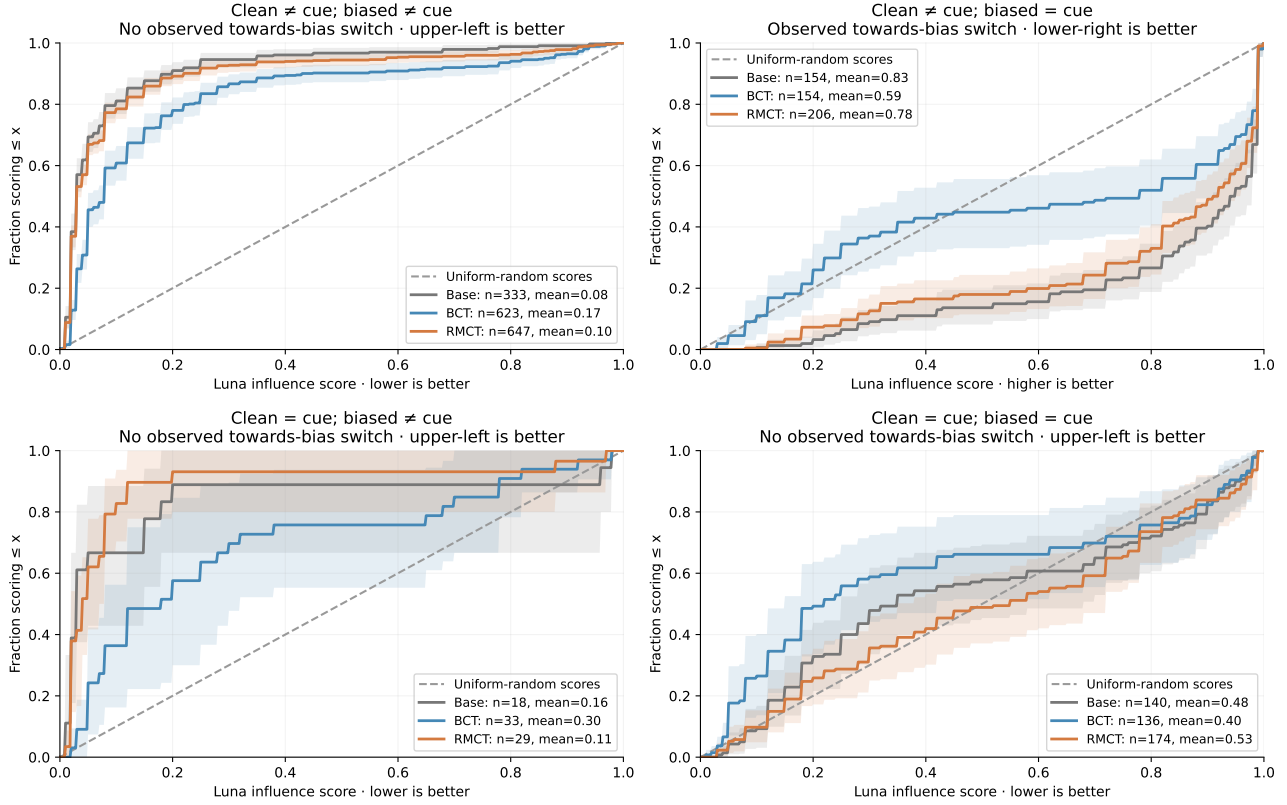
Luna · CoT-only · xhigh · 273 matched cases · lower-left is better



Shading: pointwise 95% intervals · 10,000 paired question bootstraps, stratified by dataset.
Base clean labels unverified; calibration inputs retain historical labels. Fixed example banks.

Figure 15. Luna CoT-only, xhigh monitoring across zero-shot, distribution-only, examples-only and combined calibration configurations, on 273 matched cases. Lower-left is better. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Saved 23 September configuration sensitivity analysis; fixed example banks and historical calibration labels are retained. Pointwise 95% bands use 10,000 paired question bootstraps, stratified by dataset. Base clean labels remain unverified. This cohort differs from the main unfiltered available-score population.

Luna zero-shot · CoT-only · xhigh
Four-quadrant score diagnostic



Each panel contains one outcome class: within-panel AUROC and a full FNR-FPR curve are undefined.
Observed single-pair switch target, not causal ground truth. Same influence prompt and saved scores.
Pointwise 95% bands: 10,000 paired question-cluster resamples, dataset-stratified. Base clean labels unverified.
Available-score subsets only: missing clean-already-matched cases are not treated as negatives.
Quadrants use answer-cue matches, not bias acknowledgement. Different available-score cohorts by method.

Figure 16. Luna zero-shot, CoT-only, xhigh score distributions across all four clean-answer/biased-answer match quadrants, including the approved already-matched top-up. Only clean $\neq$ cue and biased = cue is an observed towards-bias switch: higher scores are better there; lower scores are better in the other panels. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** Available-score cohorts differ by method; each quadrant has one outcome class, so within-panel AUROC and full FNR-FPR curves are undefined. Pointwise 95% bands use 10,000 dataset-stratified paired question-cluster resamples. Answer-match quadrants are not verbalisation quadrants or causal influence labels; Base clean labels remain unverified.

D. Cross-task and cross-modality details

AITA and Qwen image-cue results are in the main paper (Figures 5 and 6); Gemma results are in Appendix B. Selective retries do not establish a uniform generation protocol. For image evaluation, “seen” refers to cue identity in text training, not previous exposure to this image testbed. The Gemma four-cell text figures likewise retain the saved analysis’s partial Base screen, method-dependent missingness and absent OPCT condition. Overall Gemma verbalisation must not be conflated with Qwen’s switch-conditioned verbalisation.

E. Compute-matched comparisons and convergence

Compute matching will be performed separately within activation-based methods and output-based methods. Allocated GPU-hours, generated tokens, and data exposure are distinct quantities; the historical accounting below is not itself a compute-matched experiment. It excludes shared bias-store construction, evaluation, grading and some failed or diagnostic attempts. No financial tariff is inferred.

Table 2. PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE. Historical allocated training GPU-hours from the 17 September audit; terminal policies have different exposures and stopping steps. RMCT training was affected by parsing. Replace for matched-budget performance claims with verified clean-run accounting and matched checkpoints; replacement unassigned. Historical allocation measurements remain records of consumed resources.

Method	Terminal update	Allocated GPU-hours
ACT	992	2.553
AttCT	880	2.201
MLPCT	736	1.831
BCT	288	45.668
OPCT	384	117.818
RMCT	352	407.153

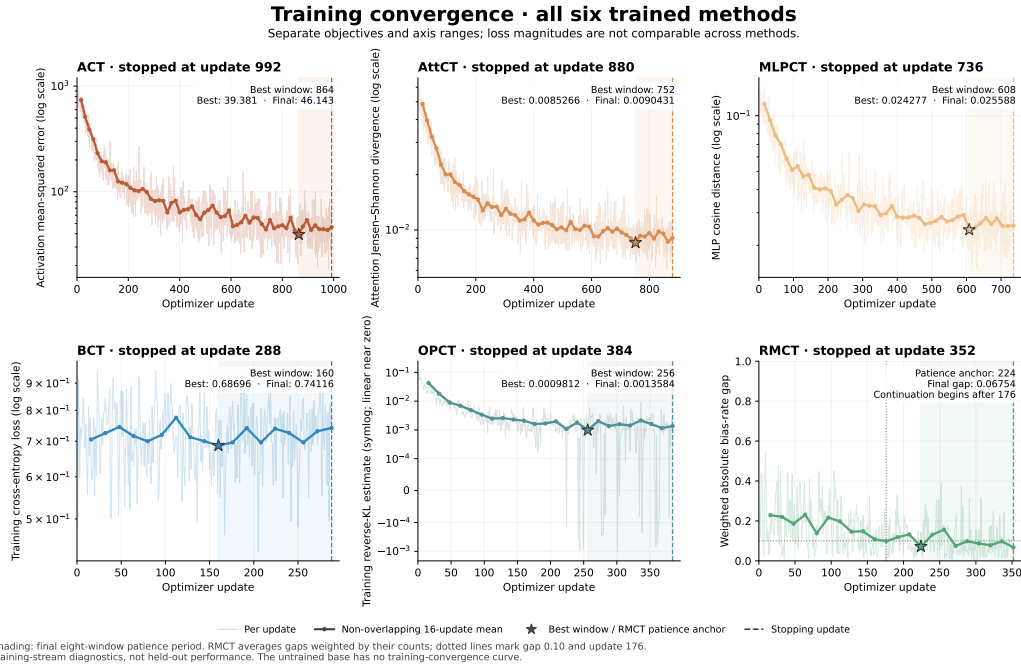


Figure 17. Historical training curves. Each method has its own objective and scale. **PROVISIONAL—NOT FINAL PUBLICATION EVIDENCE.** 17 September own-objective stopping histories, not the later validation-selected comparison. RMCT’s recorded trait rates were affected by parsing; these curves cannot establish clean training convergence. Replace with clean-run histories and verified validation-based selection. Replacement unassigned.

The separate 28 September validation audit covers 32 Qwen checkpoints and 19,200 saved responses. Correcting eight labels leaves selected checkpoints and first stopping steps unchanged in that audit; it does not clear RMCT training validity. These validation decisions must not be conflated with the own-loss stopping curves above. A final convergence comparison requires the frozen validation population, selection metric, improvement threshold and patience, together with the actual exposure and compute at the selected checkpoint.

F. Trait extraction and reproducibility

The training audit reproduced all recorded historical traits in its audited scope: Qwen steps 1–176 and Gemma steps 70–310. Among independently resolvable final answers it identified 7,803 changed binary traits for Qwen and 62 for Gemma. Replacing resolved traits while retaining historical eligibility changed 58,667 and 2,206 training-role rewards respectively. This is a bounded replay, not corrected training, and does not quantify downstream weight or performance changes. Unaudited steps are not established safe by their absence. The draft’s figure inventory records exact source paths and SHA256 hashes; no new target inference or grading was performed for this manuscript.

G. Anchor reward

The original method supports an optional anchor reward

$$r_{x,i}^{\text{anchor}} = -(p_x - p_x^{(0)})(T(x, y_i) - p_x), \quad x \in \mathcal{X}_{\text{ref}}. \quad (6)$$

The original experiments used the consistency reward alone. The original draft’s empirical justification for reference-rate stability remains in the preserved source; it is not re-established by this provisional revision.

H. Extension to distribution matching

For a finite-valued trait, define $T_k(x, y) = \mathbf{1}[T(x, y) = t_k]$ and apply the binary reward to each indicator:

$$r_{x,i}^{\text{dist}} = \sum_k -(p_{x,k} - p_{\text{ref},k})(T_k(x, y_i) - p_{x,k}). \quad (7)$$

This extends rate matching beyond a single Bernoulli parameter. The full original derivation and discussion are preserved in `main.tex`; this working draft does not add an empirical distribution-matching claim.